Algebra 2 — Unit 3: Polynomial Functions

Practice Test — Study Copy — Answer Key

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key. Write each final answer in its box. Only what is inside a box is graded.

Part A — Vocabulary (8 pts)

Write the letter of the best description in the box next to each term.

1. Degree [B]
2. Leading coefficient [A]
3. End behavior [E]
4. Multiplicity of a zero [C]
5. Turning point [D]
6. Factor Theorem [F]
7. Rational Root Theorem [G]
8. Fundamental Theorem of Algebra [H]

(A) The number multiplying the highest-power term; its sign decides the right-hand arm.

(B) The greatest exponent on the variable once the polynomial is in standard form.

(C) How many times the factor $(x-r)$ appears — odd means the graph crosses at r, even means it touches.

(D) A point where the graph changes from increasing to decreasing, or the reverse.

(E) What the graph does at its far left and far right, as $x \to \pm\infty$.

(F) $(x-r)$ is a factor of $f(x)$ exactly when $f(r)=0$.

(G) Every rational zero of a polynomial with integer coefficients looks like $\pm\frac{p}{q}$, where p divides the constant term and q divides the leading coefficient.

(H) A polynomial of degree $n \geq 1$ has exactly n complex zeros, counted with multiplicity.

Part B — Multiple Choice (12 pts)

Write the letter of the best answer in its box.

[B] 1. The end behavior of $f(x) = -2x^4 + 3x^2 - 1$ is (A) both arms up **(B) both arms down** ← (C) left down, right up (D) left up, right down

[C] 2. $(2x-5)^2 =$ (A) $4x^2 - 25$ (B) $4x^2 + 25$ **(C)** $4x^2 - 20x + 25$ ← (D) $4x^2 - 10x + 25$

[A] 3. Factored completely, $x^3 + 27$ is **(A)** $(x+3)(x^2-3x+9)$ ← (B) $(x+3)(x^2+3x+9)$ (C) $(x+3)^3$ (D) prime

[A] 4. The remainder when $f(x) = x^3 - 4x^2 + 1$ is divided by $x-2$ is **(A)** -7 ← (B) -23 (C) 7 (D) 0

C 5. Which value is *not* a possible rational zero of $f(x) = 3x^3 + 2x^2 - x + 4$? (A) -2 (B) $\frac{4}{3}$ **(C) $\frac{3}{4}$** ← (D) 1

A 6. A polynomial with *real* coefficients has degree 4 and zeros 2, -1, and $3i$. Its fourth zero must be **(A) $-3i$** ← (B) $3i$ (C) -3 (D) $\frac{1}{3}$

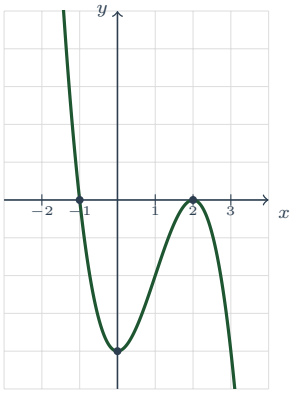
Teacher Note

Item 1: even degree ⇒ the arms match; $a = -2 < 0$ ⇒ they both point down. Only two facts are needed — the $3x^2$ and the -1 are irrelevant. Item 2: (A) and (B) are the $(a-b)^2 = a^2 - b^2$ error, (D) forgets to double the middle product. Item 4: Remainder Theorem, $f(2) = 8 - 16 + 1 = -7$; (B) is $f(-2)$ — the sign-flip error. Item 5: $p \mid 4$ and $q \mid 3$, so the list is $\pm1, \pm2, \pm4, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}$. $\frac{3}{4}$ is the *flipped* fraction. Item 6: real coefficients force conjugate pairs; $3i$ cannot travel alone.

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The polynomial function f is graphed below. Each gridline is one unit.



(a) Degree: even or odd? [odd] Leading coefficient: positive or negative? [negative]

(b) As $x \to -\infty$, $f(x) \to$ [$+\infty$]; as $x \to +\infty$, $f(x) \to$ [$-\infty$]

(c) Zeros, each with its multiplicity, and whether the graph crosses or touches there: [$x = -1$ (multiplicity 1, crosses); $x = 2$ (multiplicity 2, touches)]

(d) y-intercept: [(0,-4)]; number of turning points: [2]

(e) Relative maximum: [(2,0)]; relative minimum: [(0,-4)];
absolute maximum or minimum: [none — odd degree, so the arms run to $\pm\infty$]

(f) Interval(s) where f is increasing: [(0,2)]; domain: [all reals]; range: [all reals]

2. **Operations.** Simplify. Write each answer in standard form.

(a) $(3x^3 - 5x + 2) - (x^3 + 4x^2 - 5) = 3x^3 - 5x + 2 - x^3 - 4x^2 + 5 = 2x^3 - 4x^2 - 5x + 7$. **(The subtraction hits *every* term of the second polynomial: -5 becomes $+5$.)**

Answer: [$2x^3 - 4x^2 - 5x + 7$]

(b) $(2x - 3)(x^2 + 4x - 1) = 2x^3 + 8x^2 - 2x - 3x^2 - 12x + 3 = 2x^3 + 5x^2 - 14x + 3$

Answer: [$2x^3 + 5x^2 - 14x + 3$]

(c) For your answer to (b), give the degree [3], the leading coefficient [2], and the end behavior [left down, right up]

3. **Factor completely.** Every factor must be prime over the integers.

(a) $2x^3 - 18x$ (b) $27x^3 - 8$ (c) $x^3 + 3x^2 - 4x - 12$

(a) GCF first: $2x(x^2 - 9) = 2x(x - 3)(x + 3)$.

(b) Difference of cubes with $a = 3x$, $b = 2$: $(3x - 2)\big((3x)^2 + (3x)(2) + 2^2\big) = (3x - 2)(9x^2 + 6x + 4)$**; the quadratic factor is prime.**

(c) Four terms $\Rightarrow$ grouping: $x^2(x + 3) - 4(x + 3) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

(a) $2x(x-3)(x+3)$ **(b)** $(3x-2)(9x^2+6x+4)$ **(c)** $(x+3)(x-2)(x+2)$

4. **Division and the Remainder Theorem.**

(a) Divide using *long division* and write the answer as $Q(x) + \dfrac{R}{D(x)}$: **Answer:** $x^2+2x+6+\frac{3}{x-2}$

$$(x^3 + 2x - 9) \div (x - 2)$$

Write the dividend with a placeholder: $x^3 + 0x^2 + 2x - 9$**. Long division gives** $Q(x) = x^2 + 2x + 6$ **with** $R = 3$**, so the answer is** $x^2 + 2x + 6 + \dfrac{3}{x - 2}$**. Check:** $(x-2)(x^2+2x+6)+3 = x^3+2x-12+3 = x^3+2x-9$ ✓

(b) Use the *Remainder Theorem* to find the remainder when $g(x) = x^3 + 2x^2 - 5x + 6$ is divided by $x + 3$. Is $x + 3$ a factor of g? Explain in one sentence.

$x + 3 = x - (-3)$, so the remainder is $g(-3) = -27 + 18 + 15 + 6 = 12$. Since the remainder is 12 and not 0, $x + 3$ is *not* a factor (the Factor Theorem requires $g(-3) = 0$).

5. **Synthetic division.** You are told that $x = 3$ is a zero of

$$f(x) = x^3 - 2x^2 - 5x + 6.$$

Use synthetic division to divide out that zero, factor f completely, and list *all* of its zeros. **Synthetic division by** $r = 3$ **on** $1,\ -2,\ -5,\ 6$ **gives the bottom row** $1,\ 1,\ -2,\ 0$ **— remainder 0, confirming the zero. The depressed polynomial is** $x^2 + x - 2 = (x + 2)(x - 1)$**, so** $f(x) = (x - 3)(x + 2)(x - 1)$ **and the zeros are** $x = 3,\ x = -2,\ x = 1$**.**

Factored: $(x-3)(x+2)(x-1)$ **Zeros:** $3,\ -2,\ 1$

6. **Rational Root Theorem.** For $f(x) = 2x^3 - x^2 - 8x + 4$:

(a) List every possible rational zero. $p \mid 4$ **gives** $p = \pm1, \pm2, \pm4$**;** $q \mid 2$ **gives** $q = \pm1, \pm2$**. The list (reduced, no repeats) is** $\pm1,\ \pm2,\ \pm4,\ \pm\frac{1}{2}$**.**

Answer: $\pm1,\ \pm2,\ \pm4,\ \pm\frac{1}{2}$

(b) Find all real zeros of f, showing the test and the division you used. **Test** $x = 2$**:** $f(2) = 16 - 4 - 16 + 4 = 0$**, so** $x = 2$ **is a zero. Synthetic division by 2 on** $2,\ -1,\ -8,\ 4$ **gives** $2,\ 3,\ -2,\ 0$**, so the depressed polynomial is** $2x^2 + 3x - 2 = (2x - 1)(x + 2)$**. Hence** $f(x) = (x - 2)(2x - 1)(x + 2)$ **and the zeros are** $x = 2,\ x = \frac{1}{2},\ x = -2$**. Note** $(2x - 1)$ **gives** $x = \frac{1}{2}$ **— not 2.**

Zeros: $2,\ \frac{1}{2},\ -2$

7. **Solve over the complex numbers.** $x^3 - 3x^2 + 4x - 12 = 0$

(a) Before solving: how many solutions must this equation have, counted with multiplicity, and which theorem says so?

Exactly 3, by the Fundamental Theorem of Algebra

(b) Solve. Give every solution. **Group:** $x^2(x-3) + 4(x-3) = (x-3)(x^2+4) = 0$. **So $x = 3$, or $x^2 = -4 \Rightarrow x = \pm 2i$. The solutions are $x = 3,\ 2i,\ -2i$ — three of them, as (a) promised.**

Solutions: $3,\ 2i,\ -2i$

(c) How many of your solutions are real, and how many x-intercepts does the graph of $y = x^3 - 3x^2 + 4x - 12$ therefore have? One real solution ($x = 3$), so exactly one x-intercept — a graph can never show the imaginary pair

8. **Build a polynomial from its zeros.** Write a polynomial function of *least degree* with *real* coefficients whose zeros are -1 and $2i$. Give your answer in factored form *and* in standard form. **Real coefficients force the conjugate $-2i$ to be a zero too, so there are three zeros and the least degree is 3. Factored: $f(x) = (x+1)(x-2i)(x+2i) = (x+1)(x^2+4)$. Standard: $f(x) = x^3 + x^2 + 4x + 4$.**

Factored: $(x+1)(x^2+4)$ **Standard:** $x^3 + x^2 + 4x + 4$

Teacher Note

Item 1: (e) is the Unit 3 habit-breaker — $(2, 0)$ is a *relative* max only, and an odd-degree polynomial has *no* absolute extrema; the range is all reals for the same reason. Accept interval or inequality notation throughout. Item 1(c): full credit requires the multiplicity *and* the cross/touch, since the touch at $x = 2$ is the only evidence of the squared factor. Item 3: no credit for $(x+3)(x^2-4)$ or $2x(x^2-9)$ — "completely" is the standard. Item 4(a): the placeholder $0x^2$ is the graded step; a quotient of degree 1 means it was skipped. Item 6(b): accept a graph-based choice of which candidate to test, but the division must appear. Item 7(b): watch for $\pm$ dropped on the square root, and for "no real solutions" written as the final answer — the question says *over* $\mathbb{C}$.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The graph plan.** Consider $f(x) = -2(x+2)(x-1)^2(x-3)$. Do *not* expand it. **(a)** State the degree and the leading coefficient, and explain how you got each one from the factored form. What end behavior do they force? **(b)** List the zeros with their multiplicities, and say whether the graph crosses or touches the x-axis at each. **(c)** Find the y-intercept. **(d)** Build a sign chart and state every interval on which $f(x) > 0$. **(e)** At most how many turning points can this graph have, and why?

(a) Degrees add: $1 + 2 + 1 = 4$, so the degree is 4. Leading coefficients multiply: $-2 \cdot 1 \cdot 1 \cdot 1 = -2$. An even degree with a negative leading coefficient sends *both* arms down: $f(x) \to -\infty$ at each end.

(b) $x = -2$ (multiplicity 1, crosses); $x = 1$ (multiplicity 2, touches); $x = 3$ (multiplicity 1, crosses).

(c) $f(0) = -2(2)(-1)^2(-3) = 12$, so the y-intercept is $(0, 12)$.

(d) Test one value in each interval, keeping only signs: $f(-3) < 0$, $f(0) = 12 > 0$, $f(2) = 8 > 0$, $f(4) < 0$. So $f(x) > 0$ on $(-2, 1) \cup (1, 3)$ — that is, for $-2 < x < 3$ except $x = 1$ — and $f(x) < 0$ on $(-\infty, -2) \cup (3, \infty)$. The sign does *not* flip at $x = 1$; that is the fingerprint of the even multiplicity.

(e) At most $4 - 1 = 3$ turning points, since a degree-n polynomial turns at most $n - 1$ times.

2. **Modeling.** An open-top box is made from a 12 in by 6 in sheet of cardboard by cutting a square of side x from each corner and folding up the sides. **(a)** Write the volume $V(x)$ in factored form, then expand it into standard form. **(b)** State the feasible domain of the model and justify it from the sheet's dimensions. **(c)** Find $V(1)$ and state its units. **(d)** Check that $x = 6$ makes $V(x) = 0$. Explain why 6 is a genuine zero of the polynomial but not a possible cut length.

(a) Cutting x from both ends of each side leaves a base of $(12 - 2x)$ by $(6 - 2x)$ and a height of x, so $V(x) = x(12 - 2x)(6 - 2x)$. Expanding: $(12 - 2x)(6 - 2x) = 4x^2 - 36x + 72$, so $V(x) = 4x^3 - 36x^2 + 72x$.

(b) Every dimension must be positive: $x > 0$, $12 - 2x > 0$ (so $x < 6$), and $6 - 2x > 0$ (so $x < 3$). The tightest of these wins, so the feasible domain is $0 < x < 3$ — the short side is what limits the cut.

(c) $V(1) = 1(10)(4) = 40$ cubic inches (in^3). Check against standard form: $4 - 36 + 72 = 40$ ✓

(d) $V(6) = 6(12 - 12)(6 - 12) = 6 \cdot 0 \cdot (-6) = 0$, so 6 really is a zero of the polynomial. But a 6-inch cut from a 6-inch side removes the entire width, and $6 - 2(6) = -6$ is a negative length — there is no box at all. The polynomial does not know about cardboard; only the feasible domain $0 < x < 3$ describes a real box.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) degree 4 *and* lead -2 with both arms down; 2 pts (b) all three zeros with multiplicities and cross/touch; 1 pt (c) $(0, 12)$; 1 pt (d) $(-2, 1) \cup (1, 3)$ with the no-flip at $x = 1$ noted; 1 pt (e) 3, justified by $n - 1$. Give no credit in (a) for "degree 3" from counting factors — the exponents must be added. D2: 2 pts (a) factored *and* expanded; 2 pts (b) $0 < x < 3$ justified by $6 - 2x > 0$ (a domain of $0 < x < 6$ earns 1); 1 pt (c) 40 in^3 with cubic units; 1 pt (d) $x = 6$ recognized as a real zero of V but rejected because the width $6 - 2x$ would be negative. Do not accept "because it is too big" without the dimension.